

# NCERT CLASS 5

## 60 Questions Per Topic

Based on the refined 2026–27 curriculum map created for this project.

Each curriculum topic below contains exactly 60 questions: understanding, examples/classification, comparison, application, reasoning/error analysis, and higher-order activity.

These are question-bank items derived from the topic descriptions in the supplied curriculum map. Validate against the exact school/textbook edition before high-stakes use.

# MATHEMATICS — 1. We the Travellers-I

Coverage: ■ Numbers, travel contexts, distance, estimation and mathematical communication.

## Understanding

1. What is ■ Numbers, and what are its main features? Use a simple example.
2. What is travel contexts, and what are its main features? Use a school-life example.
3. What is distance, and what are its main features? Use a real-world example.
4. What is estimation and mathematical communication., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Numbers, and what are its main features? Explain in your own words.
6. What is travel contexts, and what are its main features? Give a step-by-step response.
7. What is distance, and what are its main features? Mention one common misconception.
8. What is estimation and mathematical communication., and what are its main features? State one practical application.
9. What is ■ Numbers, and what are its main features? Justify your response.
10. What is travel contexts, and what are its main features? Include the relevant rule or principle.

## Examples & Classification

11. Give two examples related to distance and explain each. Use a simple example.
12. Give two examples related to estimation and mathematical communication. and explain each. Use a school-life example.
13. Give two examples related to ■ Numbers and explain each. Use a real-world example.
14. Give two examples related to travel contexts and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to distance and explain each. Explain in your own words.
16. Give two examples related to estimation and mathematical communication. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Numbers and explain each. Mention one common misconception.
18. Give two examples related to travel contexts and explain each. State one practical application.
19. Give two examples related to distance and explain each. Justify your response.
20. Give two examples related to estimation and mathematical communication. and explain each. Include the relevant rule or principle.

## Comparison

21. How is ■ Numbers related to travel contexts? Compare the two ideas. Use a simple example.
22. How is travel contexts related to distance? Compare the two ideas. Use a school-life example.
23. How is distance related to estimation and mathematical communication.? Compare the two ideas. Use a real-world example.
24. How is estimation and mathematical communication. related to ■ Numbers? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Numbers related to travel contexts? Compare the two ideas. Explain in your own words.
26. How is travel contexts related to distance? Compare the two ideas. Give a step-by-step response.
27. How is distance related to estimation and mathematical communication.? Compare the two ideas. Mention one common misconception.
28. How is estimation and mathematical communication. related to ■ Numbers? Compare the two ideas. State one practical application.
29. How is ■ Numbers related to travel contexts? Compare the two ideas. Justify your response.
30. How is travel contexts related to distance? Compare the two ideas. Include the relevant rule or principle.

## Application

31. Apply distance to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply estimation and mathematical communication. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Numbers to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply travel contexts to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply distance to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply estimation and mathematical communication. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Numbers to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply travel contexts to a realistic situation. What would you do or calculate? State one practical application.
39. Apply distance to a realistic situation. What would you do or calculate? Justify your response.
40. Apply estimation and mathematical communication. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

## Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Numbers. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of travel contexts. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of distance. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of estimation and mathematical communication.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Numbers. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of travel contexts. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of distance. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of estimation and mathematical communication.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Numbers. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of travel contexts. Identify a likely error and explain the correction. Include the relevant rule or principle.

## Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show distance. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show estimation and mathematical communication.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Numbers. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show travel contexts. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show distance. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show estimation and mathematical communication.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Numbers. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show travel contexts. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show distance. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show estimation and mathematical communication.. Include the relevant rule or principle.

# MATHEMATICS — 2. Fractions

Coverage: ■ Fractions as parts/quantities; equivalent fractions; comparison; operations in simple contexts.

## Understanding

1. What is ■ Fractions as parts/quantities, and what are its main features? Use a simple example.
2. What is equivalent fractions, and what are its main features? Use a school-life example.
3. What is comparison, and what are its main features? Use a real-world example.
4. What is operations in simple contexts., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Fractions as parts/quantities, and what are its main features? Explain in your own words.
6. What is equivalent fractions, and what are its main features? Give a step-by-step response.
7. What is comparison, and what are its main features? Mention one common misconception.
8. What is operations in simple contexts., and what are its main features? State one practical application.
9. What is ■ Fractions as parts/quantities, and what are its main features? Justify your response.
10. What is equivalent fractions, and what are its main features? Include the relevant rule or principle.

## Examples & Classification

11. Give two examples related to comparison and explain each. Use a simple example.
12. Give two examples related to operations in simple contexts. and explain each. Use a school-life example.
13. Give two examples related to ■ Fractions as parts/quantities and explain each. Use a real-world example.
14. Give two examples related to equivalent fractions and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to comparison and explain each. Explain in your own words.
16. Give two examples related to operations in simple contexts. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Fractions as parts/quantities and explain each. Mention one common misconception.
18. Give two examples related to equivalent fractions and explain each. State one practical application.
19. Give two examples related to comparison and explain each. Justify your response.
20. Give two examples related to operations in simple contexts. and explain each. Include the relevant rule or principle.

## Comparison

21. How is ■ Fractions as parts/quantities related to equivalent fractions? Compare the two ideas. Use a simple example.
22. How is equivalent fractions related to comparison? Compare the two ideas. Use a school-life example.
23. How is comparison related to operations in simple contexts.? Compare the two ideas. Use a real-world example.
24. How is operations in simple contexts. related to ■ Fractions as parts/quantities? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Fractions as parts/quantities related to equivalent fractions? Compare the two ideas. Explain in your own words.
26. How is equivalent fractions related to comparison? Compare the two ideas. Give a step-by-step response.
27. How is comparison related to operations in simple contexts.? Compare the two ideas. Mention one common misconception.
28. How is operations in simple contexts. related to ■ Fractions as parts/quantities? Compare the two ideas. State one practical application.
29. How is ■ Fractions as parts/quantities related to equivalent fractions? Compare the two ideas. Justify your response.
30. How is equivalent fractions related to comparison? Compare the two ideas. Include the relevant rule or principle.

## Application

31. Apply comparison to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply operations in simple contexts. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Fractions as parts/quantities to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply equivalent fractions to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply comparison to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply operations in simple contexts. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Fractions as parts/quantities to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply equivalent fractions to a realistic situation. What would you do or calculate? State one practical application.
39. Apply comparison to a realistic situation. What would you do or calculate? Justify your response.
40. Apply operations in simple contexts. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

## Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Fractions as parts/quantities. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of equivalent fractions. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of comparison. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of operations in simple contexts.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Fractions as parts/quantities. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of equivalent fractions. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of comparison. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of operations in simple contexts.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Fractions as parts/quantities. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of equivalent fractions. Identify a likely error and explain the correction. Include the relevant rule or principle.

## Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show comparison. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show operations in simple contexts.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Fractions as parts/quantities. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show equivalent fractions. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show comparison. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show operations in simple contexts.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Fractions as parts/quantities. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show equivalent fractions. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show comparison. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show operations in simple contexts.. Include the relevant rule or principle.

# MATHEMATICS — 3. Angles as Turns

Coverage: ■ Turns, angles, right/acute/obtuse ideas; angle comparison.

## Understanding

1. What is ■ Turns, and what are its main features? Use a simple example.
2. What is angles, and what are its main features? Use a school-life example.
3. What is right/acute/obtuse ideas, and what are its main features? Use a real-world example.
4. What is angle comparison., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Turns, and what are its main features? Explain in your own words.
6. What is angles, and what are its main features? Give a step-by-step response.
7. What is right/acute/obtuse ideas, and what are its main features? Mention one common misconception.
8. What is angle comparison., and what are its main features? State one practical application.
9. What is ■ Turns, and what are its main features? Justify your response.
10. What is angles, and what are its main features? Include the relevant rule or principle.

## Examples & Classification

11. Give two examples related to right/acute/obtuse ideas and explain each. Use a simple example.
12. Give two examples related to angle comparison. and explain each. Use a school-life example.
13. Give two examples related to ■ Turns and explain each. Use a real-world example.
14. Give two examples related to angles and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to right/acute/obtuse ideas and explain each. Explain in your own words.
16. Give two examples related to angle comparison. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Turns and explain each. Mention one common misconception.
18. Give two examples related to angles and explain each. State one practical application.
19. Give two examples related to right/acute/obtuse ideas and explain each. Justify your response.
20. Give two examples related to angle comparison. and explain each. Include the relevant rule or principle.

## Comparison

21. How is ■ Turns related to angles? Compare the two ideas. Use a simple example.
22. How is angles related to right/acute/obtuse ideas? Compare the two ideas. Use a school-life example.
23. How is right/acute/obtuse ideas related to angle comparison.? Compare the two ideas. Use a real-world example.
24. How is angle comparison. related to ■ Turns? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Turns related to angles? Compare the two ideas. Explain in your own words.
26. How is angles related to right/acute/obtuse ideas? Compare the two ideas. Give a step-by-step response.
27. How is right/acute/obtuse ideas related to angle comparison.? Compare the two ideas. Mention one common misconception.
28. How is angle comparison. related to ■ Turns? Compare the two ideas. State one practical application.
29. How is ■ Turns related to angles? Compare the two ideas. Justify your response.
30. How is angles related to right/acute/obtuse ideas? Compare the two ideas. Include the relevant rule or principle.

## Application

31. Apply right/acute/obtuse ideas to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply angle comparison. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Turns to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply angles to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply right/acute/obtuse ideas to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply angle comparison. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Turns to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply angles to a realistic situation. What would you do or calculate? State one practical application.
39. Apply right/acute/obtuse ideas to a realistic situation. What would you do or calculate? Justify your response.
40. Apply angle comparison. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

### Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Turns. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of angles. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of right/acute/obtuse ideas. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of angle comparison.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Turns. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of angles. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of right/acute/obtuse ideas. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of angle comparison.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Turns. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of angles. Identify a likely error and explain the correction. Include the relevant rule or principle.

### Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show right/acute/obtuse ideas. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show angle comparison.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Turns. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show angles. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show right/acute/obtuse ideas. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show angle comparison.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Turns. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show angles. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show right/acute/obtuse ideas. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show angle comparison.. Include the relevant rule or principle.

# MATHEMATICS — 4. We the Travellers-II

Coverage: ■ Distance, time, speed intuition and multi-step travel problems.

## Understanding

1. What is ■ Distance, and what are its main features? Use a simple example.
2. What is time, and what are its main features? Use a school-life example.
3. What is speed intuition and multi-step travel problems., and what are its main features? Use a real-world example.
4. What is ■ Distance, and what are its main features? Show the idea with a diagram where appropriate.
5. What is time, and what are its main features? Explain in your own words.
6. What is speed intuition and multi-step travel problems., and what are its main features? Give a step-by-step response.
7. What is ■ Distance, and what are its main features? Mention one common misconception.
8. What is time, and what are its main features? State one practical application.
9. What is speed intuition and multi-step travel problems., and what are its main features? Justify your response.
10. What is ■ Distance, and what are its main features? Include the relevant rule or principle.

## Examples & Classification

11. Give two examples related to time and explain each. Use a simple example.
12. Give two examples related to speed intuition and multi-step travel problems. and explain each. Use a school-life example.
13. Give two examples related to ■ Distance and explain each. Use a real-world example.
14. Give two examples related to time and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to speed intuition and multi-step travel problems. and explain each. Explain in your own words.
16. Give two examples related to ■ Distance and explain each. Give a step-by-step response.
17. Give two examples related to time and explain each. Mention one common misconception.
18. Give two examples related to speed intuition and multi-step travel problems. and explain each. State one practical application.
19. Give two examples related to ■ Distance and explain each. Justify your response.
20. Give two examples related to time and explain each. Include the relevant rule or principle.

## Comparison

21. How is speed intuition and multi-step travel problems. related to ■ Distance? Compare the two ideas. Use a simple example.
22. How is ■ Distance related to time? Compare the two ideas. Use a school-life example.
23. How is time related to speed intuition and multi-step travel problems.? Compare the two ideas. Use a real-world example.
24. How is speed intuition and multi-step travel problems. related to ■ Distance? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Distance related to time? Compare the two ideas. Explain in your own words.
26. How is time related to speed intuition and multi-step travel problems.? Compare the two ideas. Give a step-by-step response.
27. How is speed intuition and multi-step travel problems. related to ■ Distance? Compare the two ideas. Mention one common misconception.
28. How is ■ Distance related to time? Compare the two ideas. State one practical application.
29. How is time related to speed intuition and multi-step travel problems.? Compare the two ideas. Justify your response.
30. How is speed intuition and multi-step travel problems. related to ■ Distance? Compare the two ideas. Include the relevant rule or principle.

## Application

31. Apply ■ Distance to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply time to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply speed intuition and multi-step travel problems. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Distance to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply time to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply speed intuition and multi-step travel problems. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Distance to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply time to a realistic situation. What would you do or calculate? State one practical application.
39. Apply speed intuition and multi-step travel problems. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Distance to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

### Reasoning & Error Analysis

41. A student gives an incorrect explanation of time. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of speed intuition and multi-step travel problems.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Distance. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of time. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of speed intuition and multi-step travel problems.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Distance. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of time. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of speed intuition and multi-step travel problems.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Distance. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of time. Identify a likely error and explain the correction. Include the relevant rule or principle.

### Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show speed intuition and multi-step travel problems.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Distance. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show time. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show speed intuition and multi-step travel problems.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Distance. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show time. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show speed intuition and multi-step travel problems.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Distance. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show time. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show speed intuition and multi-step travel problems.. Include the relevant rule or principle.

# MATHEMATICS — 5. Far and Near

Coverage: ■ Distance, scale/position, estimation and spatial reasoning.

## Understanding

1. What is ■ Distance, and what are its main features? Use a simple example.
2. What is scale/position, and what are its main features? Use a school-life example.
3. What is estimation and spatial reasoning., and what are its main features? Use a real-world example.
4. What is ■ Distance, and what are its main features? Show the idea with a diagram where appropriate.
5. What is scale/position, and what are its main features? Explain in your own words.
6. What is estimation and spatial reasoning., and what are its main features? Give a step-by-step response.
7. What is ■ Distance, and what are its main features? Mention one common misconception.
8. What is scale/position, and what are its main features? State one practical application.
9. What is estimation and spatial reasoning., and what are its main features? Justify your response.
10. What is ■ Distance, and what are its main features? Include the relevant rule or principle.

## Examples & Classification

11. Give two examples related to scale/position and explain each. Use a simple example.
12. Give two examples related to estimation and spatial reasoning. and explain each. Use a school-life example.
13. Give two examples related to ■ Distance and explain each. Use a real-world example.
14. Give two examples related to scale/position and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to estimation and spatial reasoning. and explain each. Explain in your own words.
16. Give two examples related to ■ Distance and explain each. Give a step-by-step response.
17. Give two examples related to scale/position and explain each. Mention one common misconception.
18. Give two examples related to estimation and spatial reasoning. and explain each. State one practical application.
19. Give two examples related to ■ Distance and explain each. Justify your response.
20. Give two examples related to scale/position and explain each. Include the relevant rule or principle.

## Comparison

21. How is estimation and spatial reasoning. related to ■ Distance? Compare the two ideas. Use a simple example.
22. How is ■ Distance related to scale/position? Compare the two ideas. Use a school-life example.
23. How is scale/position related to estimation and spatial reasoning.? Compare the two ideas. Use a real-world example.
24. How is estimation and spatial reasoning. related to ■ Distance? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Distance related to scale/position? Compare the two ideas. Explain in your own words.
26. How is scale/position related to estimation and spatial reasoning.? Compare the two ideas. Give a step-by-step response.
27. How is estimation and spatial reasoning. related to ■ Distance? Compare the two ideas. Mention one common misconception.
28. How is ■ Distance related to scale/position? Compare the two ideas. State one practical application.
29. How is scale/position related to estimation and spatial reasoning.? Compare the two ideas. Justify your response.
30. How is estimation and spatial reasoning. related to ■ Distance? Compare the two ideas. Include the relevant rule or principle.

## Application

31. Apply ■ Distance to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply scale/position to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply estimation and spatial reasoning. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Distance to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply scale/position to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply estimation and spatial reasoning. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Distance to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply scale/position to a realistic situation. What would you do or calculate? State one practical application.
39. Apply estimation and spatial reasoning. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Distance to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

## Reasoning & Error Analysis

41. A student gives an incorrect explanation of scale/position. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of estimation and spatial reasoning.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Distance. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of scale/position. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of estimation and spatial reasoning.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Distance. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of scale/position. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of estimation and spatial reasoning.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Distance. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of scale/position. Identify a likely error and explain the correction. Include the relevant rule or principle.

## Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show estimation and spatial reasoning.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Distance. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show scale/position. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show estimation and spatial reasoning.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Distance. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show scale/position. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show estimation and spatial reasoning.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Distance. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show scale/position. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show estimation and spatial reasoning.. Include the relevant rule or principle.

# MATHEMATICS — 6. The Dairy Farm

Coverage: ■ Multiplication/division, fractions, measurement and data in a real context.

## Understanding

1. What is ■ Multiplication/division, and what are its main features? Use a simple example.
2. What is fractions, and what are its main features? Use a school-life example.
3. What is measurement and data in a real context., and what are its main features? Use a real-world example.
4. What is ■ Multiplication/division, and what are its main features? Show the idea with a diagram where appropriate.
5. What is fractions, and what are its main features? Explain in your own words.
6. What is measurement and data in a real context., and what are its main features? Give a step-by-step response.
7. What is ■ Multiplication/division, and what are its main features? Mention one common misconception.
8. What is fractions, and what are its main features? State one practical application.
9. What is measurement and data in a real context., and what are its main features? Justify your response.
10. What is ■ Multiplication/division, and what are its main features? Include the relevant rule or principle.

## Examples & Classification

11. Give two examples related to fractions and explain each. Use a simple example.
12. Give two examples related to measurement and data in a real context. and explain each. Use a school-life example.
13. Give two examples related to ■ Multiplication/division and explain each. Use a real-world example.
14. Give two examples related to fractions and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to measurement and data in a real context. and explain each. Explain in your own words.
16. Give two examples related to ■ Multiplication/division and explain each. Give a step-by-step response.
17. Give two examples related to fractions and explain each. Mention one common misconception.
18. Give two examples related to measurement and data in a real context. and explain each. State one practical application.
19. Give two examples related to ■ Multiplication/division and explain each. Justify your response.
20. Give two examples related to fractions and explain each. Include the relevant rule or principle.

## Comparison

21. How is measurement and data in a real context. related to ■ Multiplication/division? Compare the two ideas. Use a simple example.
22. How is ■ Multiplication/division related to fractions? Compare the two ideas. Use a school-life example.
23. How is fractions related to measurement and data in a real context.? Compare the two ideas. Use a real-world example.
24. How is measurement and data in a real context. related to ■ Multiplication/division? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Multiplication/division related to fractions? Compare the two ideas. Explain in your own words.
26. How is fractions related to measurement and data in a real context.? Compare the two ideas. Give a step-by-step response.
27. How is measurement and data in a real context. related to ■ Multiplication/division? Compare the two ideas. Mention one common misconception.
28. How is ■ Multiplication/division related to fractions? Compare the two ideas. State one practical application.
29. How is fractions related to measurement and data in a real context.? Compare the two ideas. Justify your response.
30. How is measurement and data in a real context. related to ■ Multiplication/division? Compare the two ideas. Include the relevant rule or principle.

## Application

31. Apply ■ Multiplication/division to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply fractions to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply measurement and data in a real context. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Multiplication/division to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.

35. Apply fractions to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply measurement and data in a real context. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Multiplication/division to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply fractions to a realistic situation. What would you do or calculate? State one practical application.
39. Apply measurement and data in a real context. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Multiplication/division to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

### Reasoning & Error Analysis

41. A student gives an incorrect explanation of fractions. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of measurement and data in a real context.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Multiplication/division. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of fractions. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of measurement and data in a real context.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Multiplication/division. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of fractions. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of measurement and data in a real context.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Multiplication/division. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of fractions. Identify a likely error and explain the correction. Include the relevant rule or principle.

### Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show measurement and data in a real context.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Multiplication/division. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show fractions. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show measurement and data in a real context.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Multiplication/division. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show fractions. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show measurement and data in a real context.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Multiplication/division. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show fractions. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show measurement and data in a real context.. Include the relevant rule or principle.

# MATHEMATICS — 7. Shapes and Patterns

Coverage: ■ Geometric properties; tessellation/patterns; symmetry.

## Understanding

1. What is ■ Geometric properties, and what are its main features? Use a simple example.
2. What is tessellation/patterns, and what are its main features? Use a school-life example.
3. What is symmetry., and what are its main features? Use a real-world example.
4. What is ■ Geometric properties, and what are its main features? Show the idea with a diagram where appropriate.
5. What is tessellation/patterns, and what are its main features? Explain in your own words.
6. What is symmetry., and what are its main features? Give a step-by-step response.
7. What is ■ Geometric properties, and what are its main features? Mention one common misconception.
8. What is tessellation/patterns, and what are its main features? State one practical application.
9. What is symmetry., and what are its main features? Justify your response.
10. What is ■ Geometric properties, and what are its main features? Include the relevant rule or principle.

## Examples & Classification

11. Give two examples related to tessellation/patterns and explain each. Use a simple example.
12. Give two examples related to symmetry. and explain each. Use a school-life example.
13. Give two examples related to ■ Geometric properties and explain each. Use a real-world example.
14. Give two examples related to tessellation/patterns and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to symmetry. and explain each. Explain in your own words.
16. Give two examples related to ■ Geometric properties and explain each. Give a step-by-step response.
17. Give two examples related to tessellation/patterns and explain each. Mention one common misconception.
18. Give two examples related to symmetry. and explain each. State one practical application.
19. Give two examples related to ■ Geometric properties and explain each. Justify your response.
20. Give two examples related to tessellation/patterns and explain each. Include the relevant rule or principle.

## Comparison

21. How is symmetry. related to ■ Geometric properties? Compare the two ideas. Use a simple example.
22. How is ■ Geometric properties related to tessellation/patterns? Compare the two ideas. Use a school-life example.
23. How is tessellation/patterns related to symmetry.? Compare the two ideas. Use a real-world example.
24. How is symmetry. related to ■ Geometric properties? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Geometric properties related to tessellation/patterns? Compare the two ideas. Explain in your own words.
26. How is tessellation/patterns related to symmetry.? Compare the two ideas. Give a step-by-step response.
27. How is symmetry. related to ■ Geometric properties? Compare the two ideas. Mention one common misconception.
28. How is ■ Geometric properties related to tessellation/patterns? Compare the two ideas. State one practical application.
29. How is tessellation/patterns related to symmetry.? Compare the two ideas. Justify your response.
30. How is symmetry. related to ■ Geometric properties? Compare the two ideas. Include the relevant rule or principle.

## Application

31. Apply ■ Geometric properties to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply tessellation/patterns to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply symmetry. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Geometric properties to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply tessellation/patterns to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply symmetry. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Geometric properties to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply tessellation/patterns to a realistic situation. What would you do or calculate? State one practical application.
39. Apply symmetry. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Geometric properties to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

## Reasoning & Error Analysis

41. A student gives an incorrect explanation of tessellation/patterns. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of symmetry.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Geometric properties. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of tessellation/patterns. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of symmetry.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Geometric properties. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of tessellation/patterns. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of symmetry.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Geometric properties. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of tessellation/patterns. Identify a likely error and explain the correction. Include the relevant rule or principle.

## Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show symmetry.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Geometric properties. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show tessellation/patterns. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show symmetry.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Geometric properties. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show tessellation/patterns. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show symmetry.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Geometric properties. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show tessellation/patterns. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show symmetry.. Include the relevant rule or principle.

# MATHEMATICS — 8. Weight and Capacity

Coverage: ■ Mass and capacity; standard units; conversions; practical estimation.

## Understanding

1. What is ■ Mass and capacity, and what are its main features? Use a simple example.
2. What is standard units, and what are its main features? Use a school-life example.
3. What is conversions, and what are its main features? Use a real-world example.
4. What is practical estimation., and what are its main features? Show the idea with a diagram where appropriate.
5. What is ■ Mass and capacity, and what are its main features? Explain in your own words.
6. What is standard units, and what are its main features? Give a step-by-step response.
7. What is conversions, and what are its main features? Mention one common misconception.
8. What is practical estimation., and what are its main features? State one practical application.
9. What is ■ Mass and capacity, and what are its main features? Justify your response.
10. What is standard units, and what are its main features? Include the relevant rule or principle.

## Examples & Classification

11. Give two examples related to conversions and explain each. Use a simple example.
12. Give two examples related to practical estimation. and explain each. Use a school-life example.
13. Give two examples related to ■ Mass and capacity and explain each. Use a real-world example.
14. Give two examples related to standard units and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to conversions and explain each. Explain in your own words.
16. Give two examples related to practical estimation. and explain each. Give a step-by-step response.
17. Give two examples related to ■ Mass and capacity and explain each. Mention one common misconception.
18. Give two examples related to standard units and explain each. State one practical application.
19. Give two examples related to conversions and explain each. Justify your response.
20. Give two examples related to practical estimation. and explain each. Include the relevant rule or principle.

## Comparison

21. How is ■ Mass and capacity related to standard units? Compare the two ideas. Use a simple example.
22. How is standard units related to conversions? Compare the two ideas. Use a school-life example.
23. How is conversions related to practical estimation.? Compare the two ideas. Use a real-world example.
24. How is practical estimation. related to ■ Mass and capacity? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Mass and capacity related to standard units? Compare the two ideas. Explain in your own words.
26. How is standard units related to conversions? Compare the two ideas. Give a step-by-step response.
27. How is conversions related to practical estimation.? Compare the two ideas. Mention one common misconception.
28. How is practical estimation. related to ■ Mass and capacity? Compare the two ideas. State one practical application.
29. How is ■ Mass and capacity related to standard units? Compare the two ideas. Justify your response.
30. How is standard units related to conversions? Compare the two ideas. Include the relevant rule or principle.

## Application

31. Apply conversions to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply practical estimation. to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply ■ Mass and capacity to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply standard units to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply conversions to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply practical estimation. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Mass and capacity to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply standard units to a realistic situation. What would you do or calculate? State one practical application.
39. Apply conversions to a realistic situation. What would you do or calculate? Justify your response.
40. Apply practical estimation. to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

### Reasoning & Error Analysis

41. A student gives an incorrect explanation of ■ Mass and capacity. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of standard units. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of conversions. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of practical estimation.. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of ■ Mass and capacity. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of standard units. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of conversions. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of practical estimation.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Mass and capacity. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of standard units. Identify a likely error and explain the correction. Include the relevant rule or principle.

### Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show conversions. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show practical estimation.. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Mass and capacity. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show standard units. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show conversions. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show practical estimation.. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Mass and capacity. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show standard units. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show conversions. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show practical estimation.. Include the relevant rule or principle.

# MATHEMATICS — 9. Coconut Farm

Coverage: ■ Multiplication, division, area/measurement and data in context.

## Understanding

1. What is ■ Multiplication, and what are its main features? Use a simple example.
2. What is division, and what are its main features? Use a school-life example.
3. What is area/measurement and data in context., and what are its main features? Use a real-world example.
4. What is ■ Multiplication, and what are its main features? Show the idea with a diagram where appropriate.
5. What is division, and what are its main features? Explain in your own words.
6. What is area/measurement and data in context., and what are its main features? Give a step-by-step response.
7. What is ■ Multiplication, and what are its main features? Mention one common misconception.
8. What is division, and what are its main features? State one practical application.
9. What is area/measurement and data in context., and what are its main features? Justify your response.
10. What is ■ Multiplication, and what are its main features? Include the relevant rule or principle.

## Examples & Classification

11. Give two examples related to division and explain each. Use a simple example.
12. Give two examples related to area/measurement and data in context. and explain each. Use a school-life example.
13. Give two examples related to ■ Multiplication and explain each. Use a real-world example.
14. Give two examples related to division and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to area/measurement and data in context. and explain each. Explain in your own words.
16. Give two examples related to ■ Multiplication and explain each. Give a step-by-step response.
17. Give two examples related to division and explain each. Mention one common misconception.
18. Give two examples related to area/measurement and data in context. and explain each. State one practical application.
19. Give two examples related to ■ Multiplication and explain each. Justify your response.
20. Give two examples related to division and explain each. Include the relevant rule or principle.

## Comparison

21. How is area/measurement and data in context. related to ■ Multiplication? Compare the two ideas. Use a simple example.
22. How is ■ Multiplication related to division? Compare the two ideas. Use a school-life example.
23. How is division related to area/measurement and data in context.? Compare the two ideas. Use a real-world example.
24. How is area/measurement and data in context. related to ■ Multiplication? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Multiplication related to division? Compare the two ideas. Explain in your own words.
26. How is division related to area/measurement and data in context.? Compare the two ideas. Give a step-by-step response.
27. How is area/measurement and data in context. related to ■ Multiplication? Compare the two ideas. Mention one common misconception.
28. How is ■ Multiplication related to division? Compare the two ideas. State one practical application.
29. How is division related to area/measurement and data in context.? Compare the two ideas. Justify your response.
30. How is area/measurement and data in context. related to ■ Multiplication? Compare the two ideas. Include the relevant rule or principle.

## Application

31. Apply ■ Multiplication to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply division to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply area/measurement and data in context. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Multiplication to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply division to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply area/measurement and data in context. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Multiplication to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply division to a realistic situation. What would you do or calculate? State one practical application.
39. Apply area/measurement and data in context. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Multiplication to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

## Reasoning & Error Analysis

41. A student gives an incorrect explanation of division. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of area/measurement and data in context.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Multiplication. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of division. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of area/measurement and data in context.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Multiplication. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of division. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of area/measurement and data in context.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Multiplication. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of division. Identify a likely error and explain the correction. Include the relevant rule or principle.

## Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show area/measurement and data in context.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Multiplication. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show division. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show area/measurement and data in context.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Multiplication. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show division. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show area/measurement and data in context.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Multiplication. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show division. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show area/measurement and data in context.. Include the relevant rule or principle.

# MATHEMATICS — 10. Symmetrical Designs

Coverage: ■ Line symmetry; reflections; geometric design.

## Understanding

1. What is ■ Line symmetry, and what are its main features? Use a simple example.
2. What is reflections, and what are its main features? Use a school-life example.
3. What is geometric design., and what are its main features? Use a real-world example.
4. What is ■ Line symmetry, and what are its main features? Show the idea with a diagram where appropriate.
5. What is reflections, and what are its main features? Explain in your own words.
6. What is geometric design., and what are its main features? Give a step-by-step response.
7. What is ■ Line symmetry, and what are its main features? Mention one common misconception.
8. What is reflections, and what are its main features? State one practical application.
9. What is geometric design., and what are its main features? Justify your response.
10. What is ■ Line symmetry, and what are its main features? Include the relevant rule or principle.

## Examples & Classification

11. Give two examples related to reflections and explain each. Use a simple example.
12. Give two examples related to geometric design. and explain each. Use a school-life example.
13. Give two examples related to ■ Line symmetry and explain each. Use a real-world example.
14. Give two examples related to reflections and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to geometric design. and explain each. Explain in your own words.
16. Give two examples related to ■ Line symmetry and explain each. Give a step-by-step response.
17. Give two examples related to reflections and explain each. Mention one common misconception.
18. Give two examples related to geometric design. and explain each. State one practical application.
19. Give two examples related to ■ Line symmetry and explain each. Justify your response.
20. Give two examples related to reflections and explain each. Include the relevant rule or principle.

## Comparison

21. How is geometric design. related to ■ Line symmetry? Compare the two ideas. Use a simple example.
22. How is ■ Line symmetry related to reflections? Compare the two ideas. Use a school-life example.
23. How is reflections related to geometric design.? Compare the two ideas. Use a real-world example.
24. How is geometric design. related to ■ Line symmetry? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Line symmetry related to reflections? Compare the two ideas. Explain in your own words.
26. How is reflections related to geometric design.? Compare the two ideas. Give a step-by-step response.
27. How is geometric design. related to ■ Line symmetry? Compare the two ideas. Mention one common misconception.
28. How is ■ Line symmetry related to reflections? Compare the two ideas. State one practical application.
29. How is reflections related to geometric design.? Compare the two ideas. Justify your response.
30. How is geometric design. related to ■ Line symmetry? Compare the two ideas. Include the relevant rule or principle.

## Application

31. Apply ■ Line symmetry to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply reflections to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply geometric design. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Line symmetry to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply reflections to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply geometric design. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Line symmetry to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply reflections to a realistic situation. What would you do or calculate? State one practical application.
39. Apply geometric design. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Line symmetry to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

### Reasoning & Error Analysis

41. A student gives an incorrect explanation of reflections. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of geometric design.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Line symmetry. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of reflections. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of geometric design.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Line symmetry. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of reflections. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of geometric design.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Line symmetry. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of reflections. Identify a likely error and explain the correction. Include the relevant rule or principle.

### Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show geometric design.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Line symmetry. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show reflections. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show geometric design.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Line symmetry. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show reflections. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show geometric design.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Line symmetry. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show reflections. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show geometric design.. Include the relevant rule or principle.

# MATHEMATICS — 11. Grandmother's Quilt

Coverage: ■ Patterns, area, fractions and geometric reasoning.

## Understanding

1. What is ■ Patterns, and what are its main features? Use a simple example.
2. What is area, and what are its main features? Use a school-life example.
3. What is fractions and geometric reasoning., and what are its main features? Use a real-world example.
4. What is ■ Patterns, and what are its main features? Show the idea with a diagram where appropriate.
5. What is area, and what are its main features? Explain in your own words.
6. What is fractions and geometric reasoning., and what are its main features? Give a step-by-step response.
7. What is ■ Patterns, and what are its main features? Mention one common misconception.
8. What is area, and what are its main features? State one practical application.
9. What is fractions and geometric reasoning., and what are its main features? Justify your response.
10. What is ■ Patterns, and what are its main features? Include the relevant rule or principle.

## Examples & Classification

11. Give two examples related to area and explain each. Use a simple example.
12. Give two examples related to fractions and geometric reasoning. and explain each. Use a school-life example.
13. Give two examples related to ■ Patterns and explain each. Use a real-world example.
14. Give two examples related to area and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to fractions and geometric reasoning. and explain each. Explain in your own words.
16. Give two examples related to ■ Patterns and explain each. Give a step-by-step response.
17. Give two examples related to area and explain each. Mention one common misconception.
18. Give two examples related to fractions and geometric reasoning. and explain each. State one practical application.
19. Give two examples related to ■ Patterns and explain each. Justify your response.
20. Give two examples related to area and explain each. Include the relevant rule or principle.

## Comparison

21. How is fractions and geometric reasoning. related to ■ Patterns? Compare the two ideas. Use a simple example.
22. How is ■ Patterns related to area? Compare the two ideas. Use a school-life example.
23. How is area related to fractions and geometric reasoning.? Compare the two ideas. Use a real-world example.
24. How is fractions and geometric reasoning. related to ■ Patterns? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Patterns related to area? Compare the two ideas. Explain in your own words.
26. How is area related to fractions and geometric reasoning.? Compare the two ideas. Give a step-by-step response.
27. How is fractions and geometric reasoning. related to ■ Patterns? Compare the two ideas. Mention one common misconception.
28. How is ■ Patterns related to area? Compare the two ideas. State one practical application.
29. How is area related to fractions and geometric reasoning.? Compare the two ideas. Justify your response.
30. How is fractions and geometric reasoning. related to ■ Patterns? Compare the two ideas. Include the relevant rule or principle.

## Application

31. Apply ■ Patterns to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply area to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply fractions and geometric reasoning. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Patterns to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply area to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply fractions and geometric reasoning. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Patterns to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply area to a realistic situation. What would you do or calculate? State one practical application.
39. Apply fractions and geometric reasoning. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Patterns to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

## Reasoning & Error Analysis

41. A student gives an incorrect explanation of area. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of fractions and geometric reasoning.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Patterns. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of area. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of fractions and geometric reasoning.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Patterns. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of area. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of fractions and geometric reasoning.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Patterns. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of area. Identify a likely error and explain the correction. Include the relevant rule or principle.

## Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show fractions and geometric reasoning.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Patterns. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show area. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show fractions and geometric reasoning.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Patterns. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show area. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show fractions and geometric reasoning.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Patterns. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show area. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show fractions and geometric reasoning.. Include the relevant rule or principle.

# MATHEMATICS — 12. Racing Seconds

Coverage: ■ Time, duration, measurement and comparison.

## Understanding

1. What is ■ Time, and what are its main features? Use a simple example.
2. What is duration, and what are its main features? Use a school-life example.
3. What is measurement and comparison., and what are its main features? Use a real-world example.
4. What is ■ Time, and what are its main features? Show the idea with a diagram where appropriate.
5. What is duration, and what are its main features? Explain in your own words.
6. What is measurement and comparison., and what are its main features? Give a step-by-step response.
7. What is ■ Time, and what are its main features? Mention one common misconception.
8. What is duration, and what are its main features? State one practical application.
9. What is measurement and comparison., and what are its main features? Justify your response.
10. What is ■ Time, and what are its main features? Include the relevant rule or principle.

## Examples & Classification

11. Give two examples related to duration and explain each. Use a simple example.
12. Give two examples related to measurement and comparison. and explain each. Use a school-life example.
13. Give two examples related to ■ Time and explain each. Use a real-world example.
14. Give two examples related to duration and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to measurement and comparison. and explain each. Explain in your own words.
16. Give two examples related to ■ Time and explain each. Give a step-by-step response.
17. Give two examples related to duration and explain each. Mention one common misconception.
18. Give two examples related to measurement and comparison. and explain each. State one practical application.
19. Give two examples related to ■ Time and explain each. Justify your response.
20. Give two examples related to duration and explain each. Include the relevant rule or principle.

## Comparison

21. How is measurement and comparison. related to ■ Time? Compare the two ideas. Use a simple example.
22. How is ■ Time related to duration? Compare the two ideas. Use a school-life example.
23. How is duration related to measurement and comparison.? Compare the two ideas. Use a real-world example.
24. How is measurement and comparison. related to ■ Time? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Time related to duration? Compare the two ideas. Explain in your own words.
26. How is duration related to measurement and comparison.? Compare the two ideas. Give a step-by-step response.
27. How is measurement and comparison. related to ■ Time? Compare the two ideas. Mention one common misconception.
28. How is ■ Time related to duration? Compare the two ideas. State one practical application.
29. How is duration related to measurement and comparison.? Compare the two ideas. Justify your response.
30. How is measurement and comparison. related to ■ Time? Compare the two ideas. Include the relevant rule or principle.

## Application

31. Apply ■ Time to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply duration to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply measurement and comparison. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Time to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply duration to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply measurement and comparison. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Time to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply duration to a realistic situation. What would you do or calculate? State one practical application.
39. Apply measurement and comparison. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Time to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

### Reasoning & Error Analysis

41. A student gives an incorrect explanation of duration. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of measurement and comparison.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Time. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of duration. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of measurement and comparison.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Time. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of duration. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of measurement and comparison.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Time. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of duration. Identify a likely error and explain the correction. Include the relevant rule or principle.

### Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show measurement and comparison.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Time. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show duration. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show measurement and comparison.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Time. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show duration. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show measurement and comparison.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Time. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show duration. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show measurement and comparison.. Include the relevant rule or principle.

# MATHEMATICS — 13. Animal Jumps

Coverage: ■ Length, distance, estimation and data.

## Understanding

1. What is ■ Length, and what are its main features? Use a simple example.
2. What is distance, and what are its main features? Use a school-life example.
3. What is estimation and data., and what are its main features? Use a real-world example.
4. What is ■ Length, and what are its main features? Show the idea with a diagram where appropriate.
5. What is distance, and what are its main features? Explain in your own words.
6. What is estimation and data., and what are its main features? Give a step-by-step response.
7. What is ■ Length, and what are its main features? Mention one common misconception.
8. What is distance, and what are its main features? State one practical application.
9. What is estimation and data., and what are its main features? Justify your response.
10. What is ■ Length, and what are its main features? Include the relevant rule or principle.

## Examples & Classification

11. Give two examples related to distance and explain each. Use a simple example.
12. Give two examples related to estimation and data. and explain each. Use a school-life example.
13. Give two examples related to ■ Length and explain each. Use a real-world example.
14. Give two examples related to distance and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to estimation and data. and explain each. Explain in your own words.
16. Give two examples related to ■ Length and explain each. Give a step-by-step response.
17. Give two examples related to distance and explain each. Mention one common misconception.
18. Give two examples related to estimation and data. and explain each. State one practical application.
19. Give two examples related to ■ Length and explain each. Justify your response.
20. Give two examples related to distance and explain each. Include the relevant rule or principle.

## Comparison

21. How is estimation and data. related to ■ Length? Compare the two ideas. Use a simple example.
22. How is ■ Length related to distance? Compare the two ideas. Use a school-life example.
23. How is distance related to estimation and data.? Compare the two ideas. Use a real-world example.
24. How is estimation and data. related to ■ Length? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Length related to distance? Compare the two ideas. Explain in your own words.
26. How is distance related to estimation and data.? Compare the two ideas. Give a step-by-step response.
27. How is estimation and data. related to ■ Length? Compare the two ideas. Mention one common misconception.
28. How is ■ Length related to distance? Compare the two ideas. State one practical application.
29. How is distance related to estimation and data.? Compare the two ideas. Justify your response.
30. How is estimation and data. related to ■ Length? Compare the two ideas. Include the relevant rule or principle.

## Application

31. Apply ■ Length to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply distance to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply estimation and data. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Length to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply distance to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply estimation and data. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Length to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply distance to a realistic situation. What would you do or calculate? State one practical application.
39. Apply estimation and data. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Length to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

## Reasoning & Error Analysis

41. A student gives an incorrect explanation of distance. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of estimation and data.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Length. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of distance. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of estimation and data.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Length. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of distance. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of estimation and data.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Length. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of distance. Identify a likely error and explain the correction. Include the relevant rule or principle.

## Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show estimation and data.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Length. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show distance. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show estimation and data.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Length. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show distance. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show estimation and data.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Length. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show distance. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show estimation and data.. Include the relevant rule or principle.

# MATHEMATICS — 14. Maps and Locations

Coverage: ■ Maps, directions, scale and coordinates-like location ideas.

## Understanding

1. What is ■ Maps, and what are its main features? Use a simple example.
2. What is directions, and what are its main features? Use a school-life example.
3. What is scale and coordinates-like location ideas., and what are its main features? Use a real-world example.
4. What is ■ Maps, and what are its main features? Show the idea with a diagram where appropriate.
5. What is directions, and what are its main features? Explain in your own words.
6. What is scale and coordinates-like location ideas., and what are its main features? Give a step-by-step response.
7. What is ■ Maps, and what are its main features? Mention one common misconception.
8. What is directions, and what are its main features? State one practical application.
9. What is scale and coordinates-like location ideas., and what are its main features? Justify your response.
10. What is ■ Maps, and what are its main features? Include the relevant rule or principle.

## Examples & Classification

11. Give two examples related to directions and explain each. Use a simple example.
12. Give two examples related to scale and coordinates-like location ideas. and explain each. Use a school-life example.
13. Give two examples related to ■ Maps and explain each. Use a real-world example.
14. Give two examples related to directions and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to scale and coordinates-like location ideas. and explain each. Explain in your own words.
16. Give two examples related to ■ Maps and explain each. Give a step-by-step response.
17. Give two examples related to directions and explain each. Mention one common misconception.
18. Give two examples related to scale and coordinates-like location ideas. and explain each. State one practical application.
19. Give two examples related to ■ Maps and explain each. Justify your response.
20. Give two examples related to directions and explain each. Include the relevant rule or principle.

## Comparison

21. How is scale and coordinates-like location ideas. related to ■ Maps? Compare the two ideas. Use a simple example.
22. How is ■ Maps related to directions? Compare the two ideas. Use a school-life example.
23. How is directions related to scale and coordinates-like location ideas.? Compare the two ideas. Use a real-world example.
24. How is scale and coordinates-like location ideas. related to ■ Maps? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Maps related to directions? Compare the two ideas. Explain in your own words.
26. How is directions related to scale and coordinates-like location ideas.? Compare the two ideas. Give a step-by-step response.
27. How is scale and coordinates-like location ideas. related to ■ Maps? Compare the two ideas. Mention one common misconception.
28. How is ■ Maps related to directions? Compare the two ideas. State one practical application.
29. How is directions related to scale and coordinates-like location ideas.? Compare the two ideas. Justify your response.
30. How is scale and coordinates-like location ideas. related to ■ Maps? Compare the two ideas. Include the relevant rule or principle.

## Application

31. Apply ■ Maps to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply directions to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply scale and coordinates-like location ideas. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Maps to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply directions to a realistic situation. What would you do or calculate? Explain in your own words.

36. Apply scale and coordinates-like location ideas. to a realistic situation. What would you do or calculate? Give a step-by-step response.
37. Apply ■ Maps to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply directions to a realistic situation. What would you do or calculate? State one practical application.
39. Apply scale and coordinates-like location ideas. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Maps to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

## Reasoning & Error Analysis

41. A student gives an incorrect explanation of directions. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of scale and coordinates-like location ideas.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Maps. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of directions. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of scale and coordinates-like location ideas.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Maps. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of directions. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of scale and coordinates-like location ideas.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Maps. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of directions. Identify a likely error and explain the correction. Include the relevant rule or principle.

## Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show scale and coordinates-like location ideas.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Maps. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show directions. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show scale and coordinates-like location ideas.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Maps. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show directions. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show scale and coordinates-like location ideas.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Maps. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show directions. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show scale and coordinates-like location ideas.. Include the relevant rule or principle.

# MATHEMATICS — 15. Data Through Pictures

Coverage: ■ Tables, pictorial graphs, interpretation and conclusions.

## Understanding

1. What is ■ Tables, and what are its main features? Use a simple example.
2. What is pictorial graphs, and what are its main features? Use a school-life example.
3. What is interpretation and conclusions., and what are its main features? Use a real-world example.
4. What is ■ Tables, and what are its main features? Show the idea with a diagram where appropriate.
5. What is pictorial graphs, and what are its main features? Explain in your own words.
6. What is interpretation and conclusions., and what are its main features? Give a step-by-step response.
7. What is ■ Tables, and what are its main features? Mention one common misconception.
8. What is pictorial graphs, and what are its main features? State one practical application.
9. What is interpretation and conclusions., and what are its main features? Justify your response.
10. What is ■ Tables, and what are its main features? Include the relevant rule or principle.

## Examples & Classification

11. Give two examples related to pictorial graphs and explain each. Use a simple example.
12. Give two examples related to interpretation and conclusions. and explain each. Use a school-life example.
13. Give two examples related to ■ Tables and explain each. Use a real-world example.
14. Give two examples related to pictorial graphs and explain each. Show the idea with a diagram where appropriate.
15. Give two examples related to interpretation and conclusions. and explain each. Explain in your own words.
16. Give two examples related to ■ Tables and explain each. Give a step-by-step response.
17. Give two examples related to pictorial graphs and explain each. Mention one common misconception.
18. Give two examples related to interpretation and conclusions. and explain each. State one practical application.
19. Give two examples related to ■ Tables and explain each. Justify your response.
20. Give two examples related to pictorial graphs and explain each. Include the relevant rule or principle.

## Comparison

21. How is interpretation and conclusions. related to ■ Tables? Compare the two ideas. Use a simple example.
22. How is ■ Tables related to pictorial graphs? Compare the two ideas. Use a school-life example.
23. How is pictorial graphs related to interpretation and conclusions.? Compare the two ideas. Use a real-world example.
24. How is interpretation and conclusions. related to ■ Tables? Compare the two ideas. Show the idea with a diagram where appropriate.
25. How is ■ Tables related to pictorial graphs? Compare the two ideas. Explain in your own words.
26. How is pictorial graphs related to interpretation and conclusions.? Compare the two ideas. Give a step-by-step response.
27. How is interpretation and conclusions. related to ■ Tables? Compare the two ideas. Mention one common misconception.
28. How is ■ Tables related to pictorial graphs? Compare the two ideas. State one practical application.
29. How is pictorial graphs related to interpretation and conclusions.? Compare the two ideas. Justify your response.
30. How is interpretation and conclusions. related to ■ Tables? Compare the two ideas. Include the relevant rule or principle.

## Application

31. Apply ■ Tables to a realistic situation. What would you do or calculate? Use a simple example.
32. Apply pictorial graphs to a realistic situation. What would you do or calculate? Use a school-life example.
33. Apply interpretation and conclusions. to a realistic situation. What would you do or calculate? Use a real-world example.
34. Apply ■ Tables to a realistic situation. What would you do or calculate? Show the idea with a diagram where appropriate.
35. Apply pictorial graphs to a realistic situation. What would you do or calculate? Explain in your own words.
36. Apply interpretation and conclusions. to a realistic situation. What would you do or calculate? Give a step-by-step response.

37. Apply ■ Tables to a realistic situation. What would you do or calculate? Mention one common misconception.
38. Apply pictorial graphs to a realistic situation. What would you do or calculate? State one practical application.
39. Apply interpretation and conclusions. to a realistic situation. What would you do or calculate? Justify your response.
40. Apply ■ Tables to a realistic situation. What would you do or calculate? Include the relevant rule or principle.

## Reasoning & Error Analysis

41. A student gives an incorrect explanation of pictorial graphs. Identify a likely error and explain the correction. Use a simple example.
42. A student gives an incorrect explanation of interpretation and conclusions.. Identify a likely error and explain the correction. Use a school-life example.
43. A student gives an incorrect explanation of ■ Tables. Identify a likely error and explain the correction. Use a real-world example.
44. A student gives an incorrect explanation of pictorial graphs. Identify a likely error and explain the correction. Show the idea with a diagram where appropriate.
45. A student gives an incorrect explanation of interpretation and conclusions.. Identify a likely error and explain the correction. Explain in your own words.
46. A student gives an incorrect explanation of ■ Tables. Identify a likely error and explain the correction. Give a step-by-step response.
47. A student gives an incorrect explanation of pictorial graphs. Identify a likely error and explain the correction. Mention one common misconception.
48. A student gives an incorrect explanation of interpretation and conclusions.. Identify a likely error and explain the correction. State one practical application.
49. A student gives an incorrect explanation of ■ Tables. Identify a likely error and explain the correction. Justify your response.
50. A student gives an incorrect explanation of pictorial graphs. Identify a likely error and explain the correction. Include the relevant rule or principle.

## Higher-Order / Activity

51. Design a short activity, investigation, problem, model, or demonstration to test or show interpretation and conclusions.. Use a simple example.
52. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Tables. Use a school-life example.
53. Design a short activity, investigation, problem, model, or demonstration to test or show pictorial graphs. Use a real-world example.
54. Design a short activity, investigation, problem, model, or demonstration to test or show interpretation and conclusions.. Show the idea with a diagram where appropriate.
55. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Tables. Explain in your own words.
56. Design a short activity, investigation, problem, model, or demonstration to test or show pictorial graphs. Give a step-by-step response.
57. Design a short activity, investigation, problem, model, or demonstration to test or show interpretation and conclusions.. Mention one common misconception.
58. Design a short activity, investigation, problem, model, or demonstration to test or show ■ Tables. State one practical application.
59. Design a short activity, investigation, problem, model, or demonstration to test or show pictorial graphs. Justify your response.
60. Design a short activity, investigation, problem, model, or demonstration to test or show interpretation and conclusions.. Include the relevant rule or principle.